

ILULISSAT JAKOBHAVN, Greenland

WMO: 042210

Lat: 69.240N

Lon: 51.066W

Elev: 102

StdP: 14.64

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.1	-16.7	-29.2	1.1	-18.5	-25.1	1.4	-13.3	36.2	24.6	32.8	23.9	3.7	150	0.917

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	60.2	46.7	57.1	45.4	54.0	44.3	48.1	55.8	46.8	53.4	45.7	51.8	15.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
44.7	43.8	47.8	43.1	41.3	46.9	41.9	39.2	46.3	19.3	55.8	18.6	53.8	18.0	52.1	53.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
30.0	26.4	23.1	DB	-22.3	65.3	9.5	2.2	-29.2	66.9	-34.7	68.2	-40.1	69.4	-47.0	71.0	
			WB	-22.5	51.1	9.1	1.7	-29.1	52.4	-34.4	53.4	-39.5	54.4	-46.2	55.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.6	10.6	6.3	6.6	18.7	32.4	42.9	47.0	44.0	36.6	25.9	19.7	15.3	(6)
	DBStd	16.98	12.10	14.17	13.78	11.06	7.09	5.65	4.72	4.26	5.58	7.63	9.94	9.81	(7)
	HDD50	8939	1220	1223	1345	939	546	223	118	192	402	748	908	1075	(8)
	HDD65	14370	1685	1643	1810	1389	1010	663	558	651	851	1212	1358	1540	(9)
	CDD50	42	0	0	0	0	1	10	24	6	1	0	0	0	(10)
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
	WSAvg	9.0	12.3	10.6	8.9	8.7	7.5	6.7	6.1	6.1	7.9	9.1	12.1	12.7	(14)
	PrecAvg	10.2	0.5	0.6	0.6	0.6	0.7	0.9	1.1	1.2	1.4	1.0	0.9	0.7	(15)
(16) Precipitation	PrecMax	15.2	1.7	1.6	1.6	1.8	2.3	3.1	3.1	3.7	6.3	3.9	3.1	2.2	(16)
	PrecMin	6.2	0.2	0.0	0.1	0.1	0.1	0.0	0.1	0.0	0.0	0.1	0.0	0.1	(17)
	PrecStd	2.3	0.4	0.5	0.4	0.4	0.5	0.7	0.8	0.9	1.3	0.8	0.8	0.5	(18)
	PrecStd	2.3	0.4	0.5	0.4	0.4	0.5	0.7	0.8	0.9	1.3	0.8	0.8	0.5	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	43.2	43.7	47.4	57.1	62.8	64.8	61.3	56.4	49.5	48.0	46.3	(19)
		MCWB	36.3	34.2	35.9	37.8	43.8	47.8	48.6	47.6	44.5	38.3	38.6	37.3	(20)
	2%	DB	37.0	37.1	35.8	41.5	51.3	58.6	60.8	57.2	51.5	44.2	43.0	40.1	(21)
		MCWB	30.7	30.1	30.9	34.3	40.9	46.0	46.9	45.7	41.5	35.5	34.8	33.2	(22)
	5%	DB	32.1	32.7	31.3	37.9	45.6	55.4	57.6	53.8	48.1	39.1	39.5	34.5	(23)
		MCWB	27.7	27.3	26.9	32.2	38.0	44.3	45.9	44.4	39.6	33.7	32.3	29.6	(24)
	10%	DB	28.0	27.8	26.4	33.8	41.3	51.9	54.7	51.5	44.3	35.4	34.3	30.0	(25)
		MCWB	24.1	23.3	23.1	30.5	36.0	42.9	45.0	43.4	38.1	31.5	29.9	25.9	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.8	34.9	36.3	38.6	44.5	49.0	50.3	50.0	45.5	40.1	39.6	37.5	(27)
		MCDB	42.8	42.0	42.1	45.5	54.8	59.0	60.2	57.3	54.4	46.4	45.9	43.4	(28)
	2%	WB	31.2	30.9	31.3	34.9	41.3	46.7	48.4	47.5	42.5	36.6	35.7	34.0	(29)
		MCDB	35.3	36.4	35.8	39.6	49.5	55.7	56.8	52.6	48.6	42.0	42.0	39.5	(30)
	5%	WB	27.5	27.3	27.0	32.8	38.8	45.1	47.2	46.1	40.9	34.3	32.9	29.9	(31)
		MCDB	31.7	32.6	30.7	36.9	44.4	53.0	54.3	50.8	45.3	38.3	37.6	34.9	(32)
	10%	WB	24.0	23.2	23.2	30.5	36.6	43.7	46.2	44.8	39.3	32.1	30.2	25.8	(33)
		MCDB	27.9	27.7	26.7	33.9	40.7	50.6	52.3	49.4	43.1	35.0	34.7	29.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	11.3	12.5	11.9	9.3	9.7	10.1	10.0	8.6	8.2	9.4	9.3	(35)
		MCDBR	15.8	16.6	18.3	15.5	15.1	16.0	15.0	14.6	12.6	11.6	13.9	13.7	(36)
		MCWBR	12.4	12.7	14.4	11.1	8.5	7.5	7.0	7.7	7.4	8.1	9.8	10.3	(37)
	5% WB	MCDBR	15.7	16.5	18.0	14.5	13.5	13.7	12.0	11.7	10.4	10.5	13.4	13.8	(38)
		MCWBR	12.8	13.1	14.7	10.9	8.3	7.4	6.9	7.4	7.0	8.1	10.1	10.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.160	0.210	0.245	0.302	0.312	0.335	0.328	0.321	0.286	0.221	N/A	N/A	(40)
	taud		1.921	2.089	2.279	2.205	2.384	2.423	2.478	2.483	2.482	2.189	N/A	N/A	(41)
	Ebn at Noon		18	182	254	265	278	273	272	260	240	174	N/A	0	(42)
	Edh at Noon		1	13	21	32	30	30	28	24	18	11	N/A	0	(43)
(44) All-Sky Solar Radiation	RadAvg		5	115	578	1250	1780	2029	1841	1233	655	206	19	0	(44)
	RadStd		0	16	50	121	156	167	176	97	46	17	2	0	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	+1.71	+6.62	N/A	+0.90	N/A	N/A	-531	-547	N/A	N/A

Nomenclature: See separate page